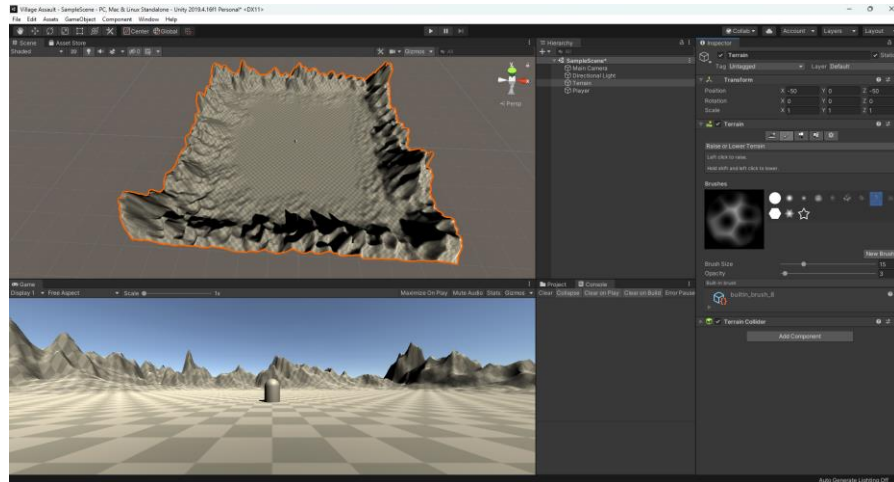


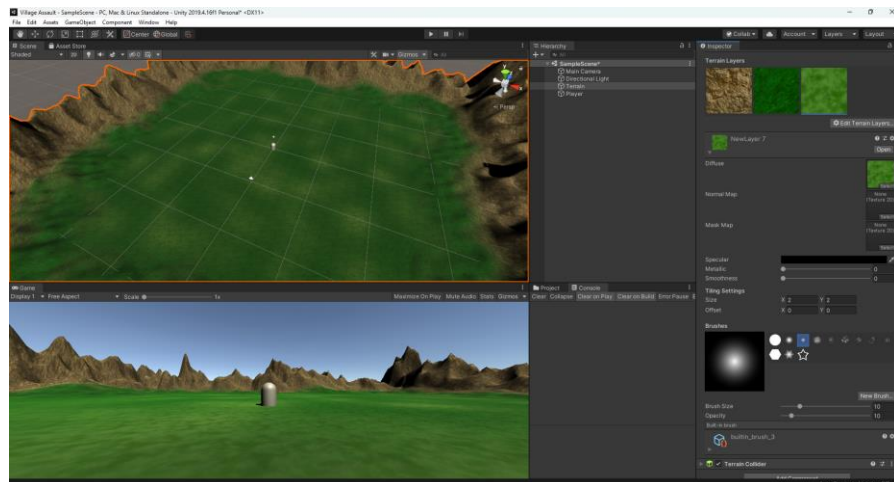
RPG Game Development Stages

Creating the Game Scene:

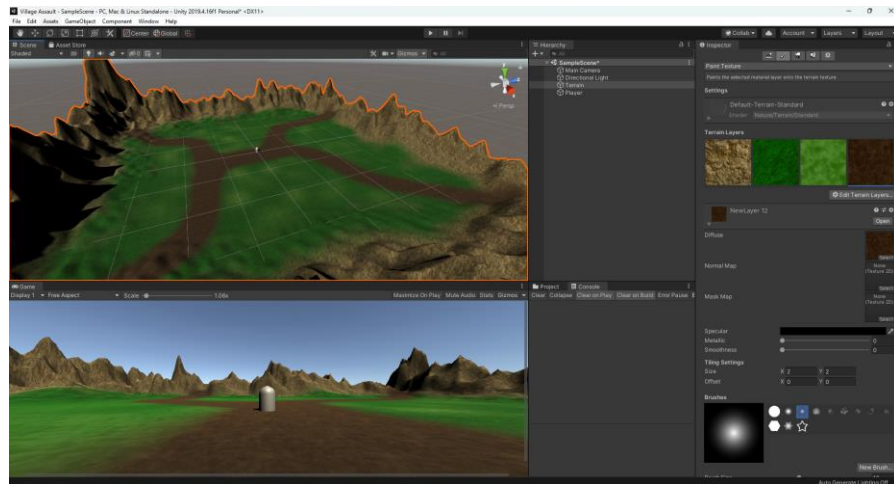
- A Terrain was created from a 3D object, followed by shaping the terrain and surrounding mountains using adjusted Brush Size and Opacity values.



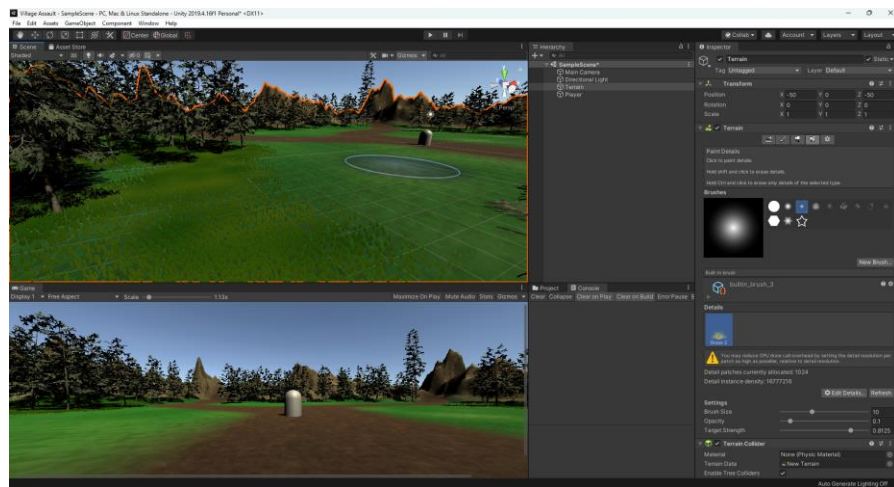
- A ready-made Assets file was imported from the internet to add textures to the 3D terrain object. These textures were applied to give the terrain a grassy appearance.



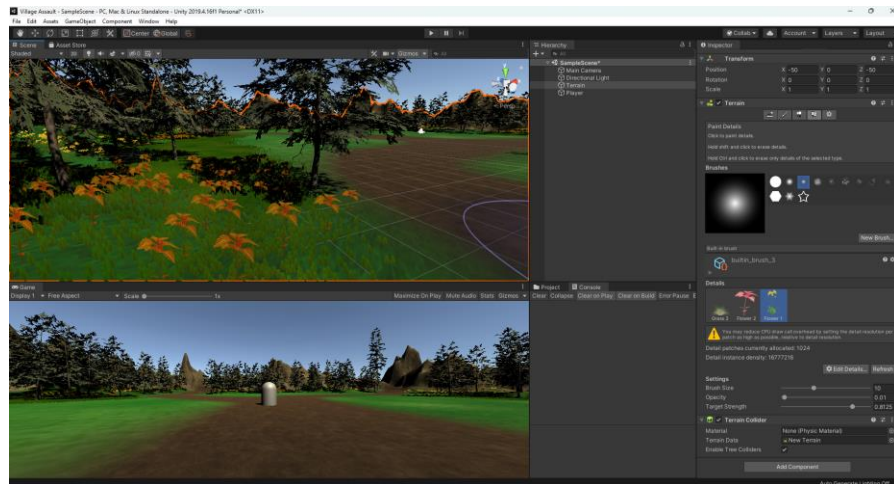
- A path was drawn using the selected texture and a specific opacity setting.



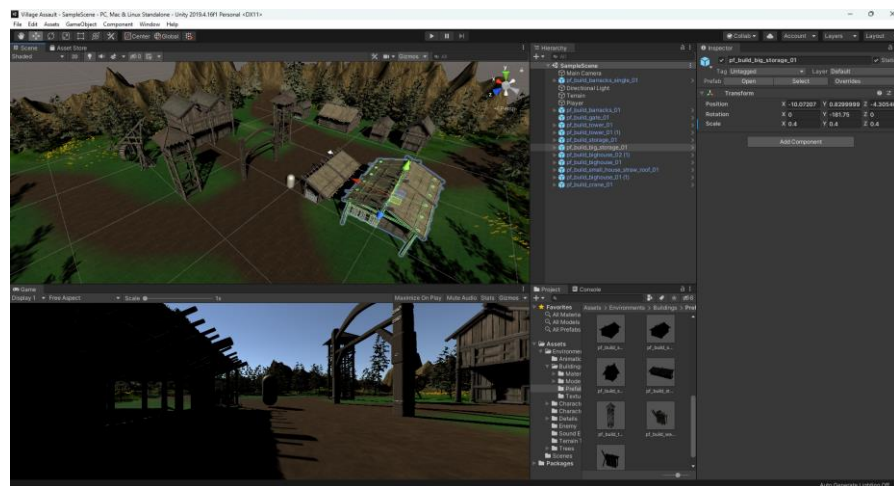
- Trees were planted using the selected tree texture. Subsequently, features such as color adjustment and density of the grass texture were set to create the area around the trees.



- Two different flower textures were added sparsely and realistically by adjusting their maximum and minimum widths and lengths, taking into account values like brush size and opacity.

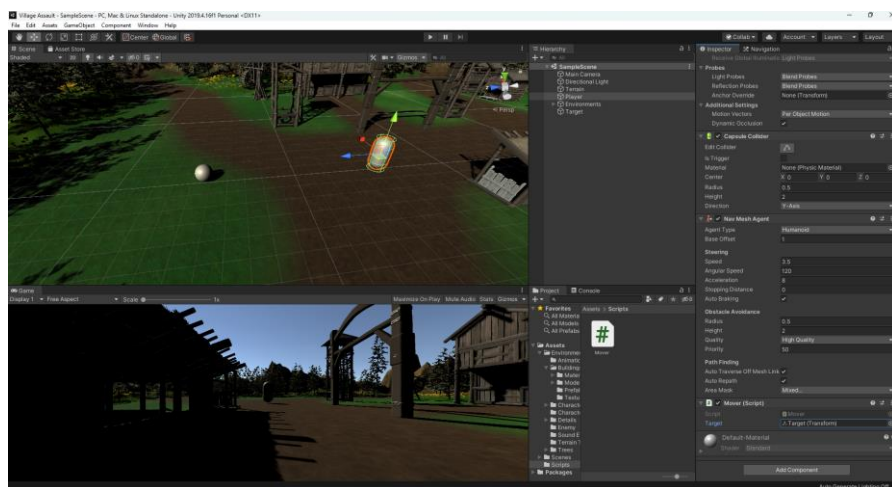
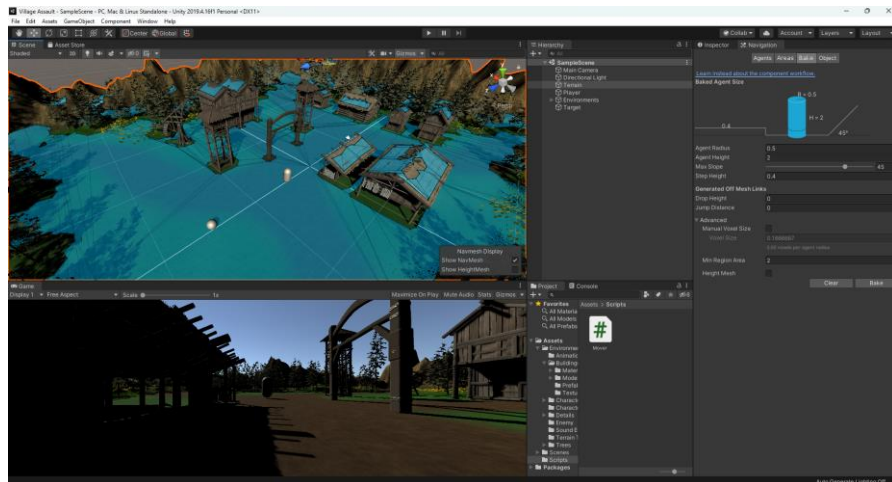


- The sizes of the ready-made building templates were adjusted and placed to create a small village.

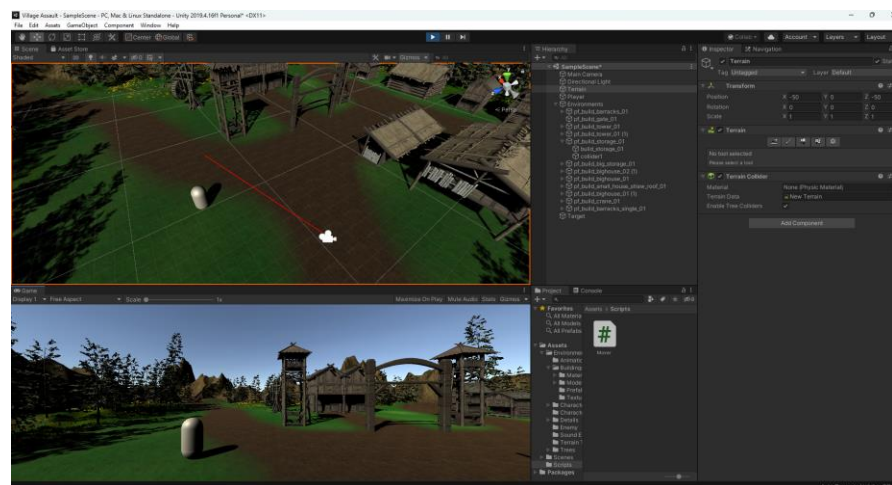


Basic Movement System:

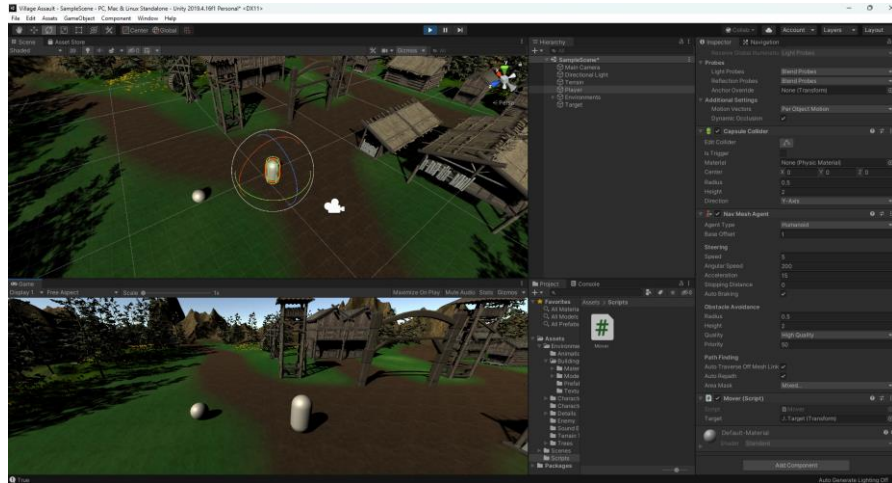
- Navigable areas were identified on the map using Unity AI Navigation. Then, with the Nav Mesh component, the Player object was coded to move towards the target object.



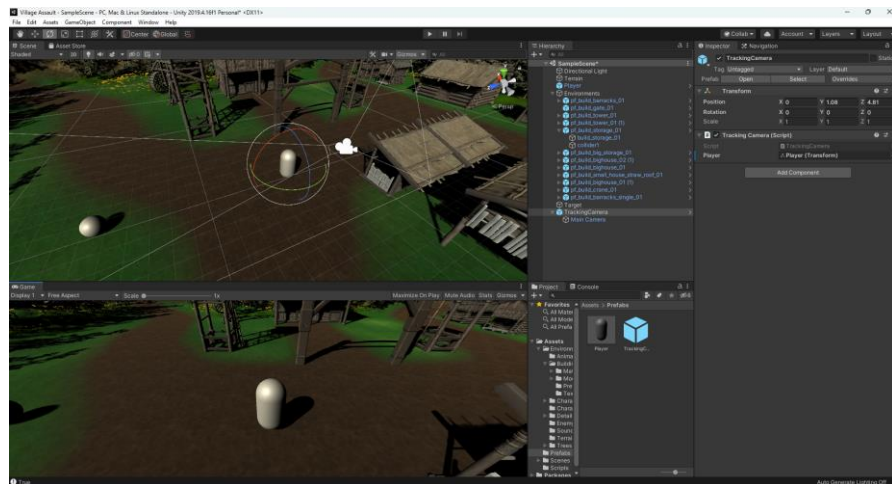
- Code was written to detect the area clicked on the game scene screen in the camera's origin region.



- With the help of code, the area clicked within the camera view is assigned to the player object, enabling it to move towards the clicked area. Movement and rotation speeds can also be set using Nav Mesh settings.

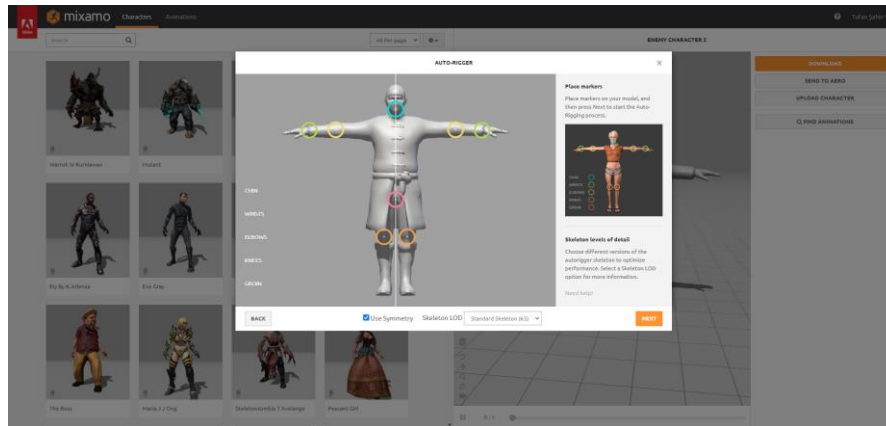


- With the necessary code, the camera object's transform values were matched with the player object's, allowing the camera to follow the player.

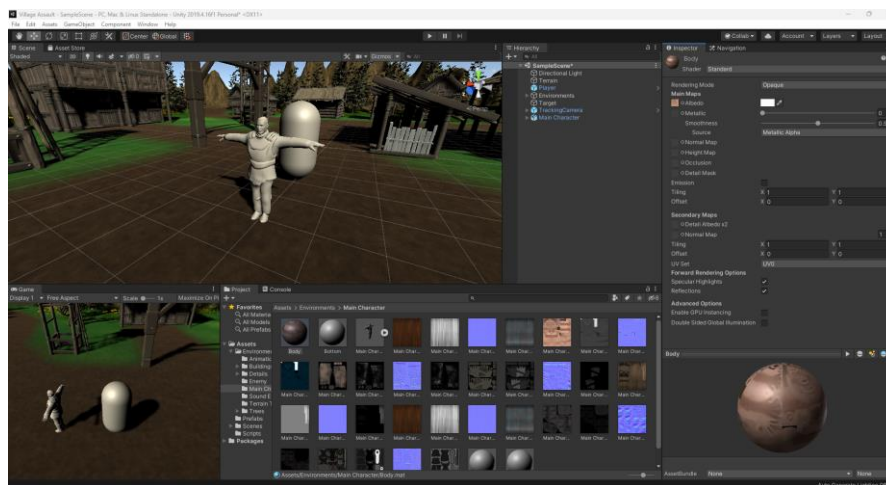


Character Creation:

- Main character and enemy designs were made with the Fuse app. Their movements were set up using the Mixamo website. Then, their textures were adjusted in Unity and added to the game.

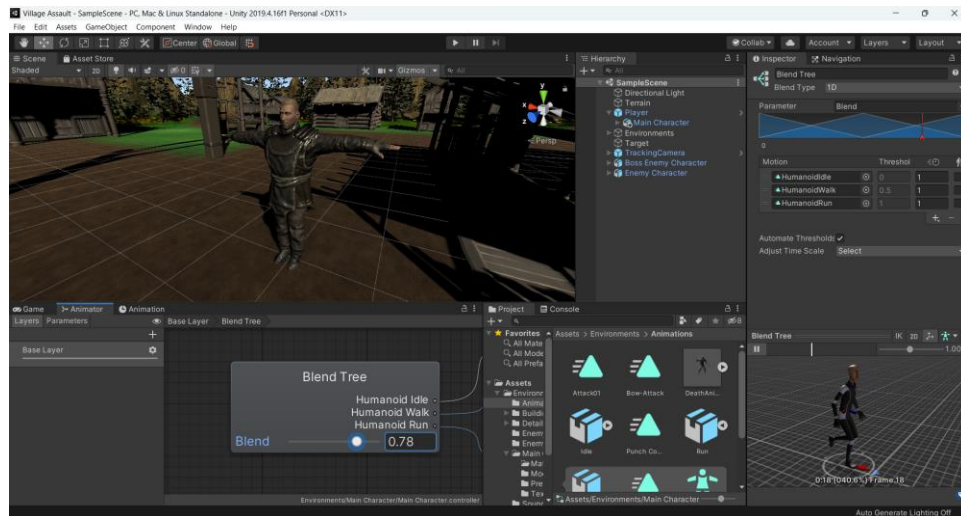


- Materials were created for the character, Albedo and Normal map values were assigned and the character was painted. The brightness of the materials was adjusted with the Smoothness setting.



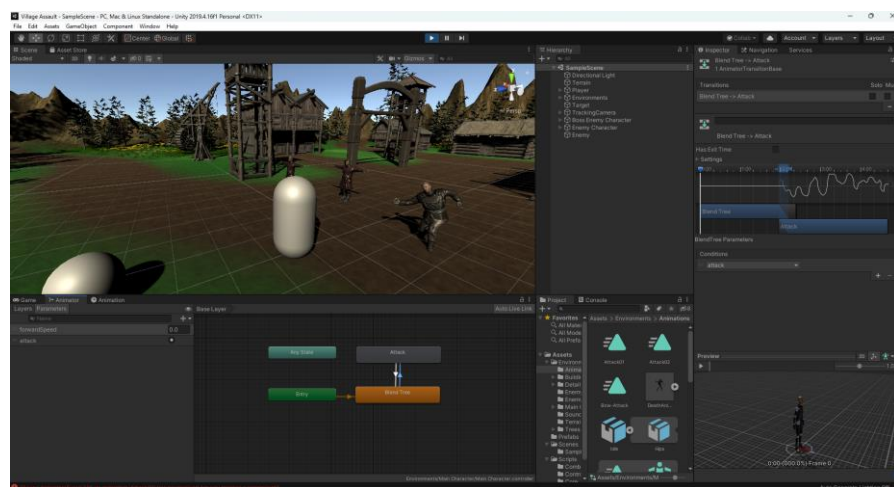
Animation System Basics:

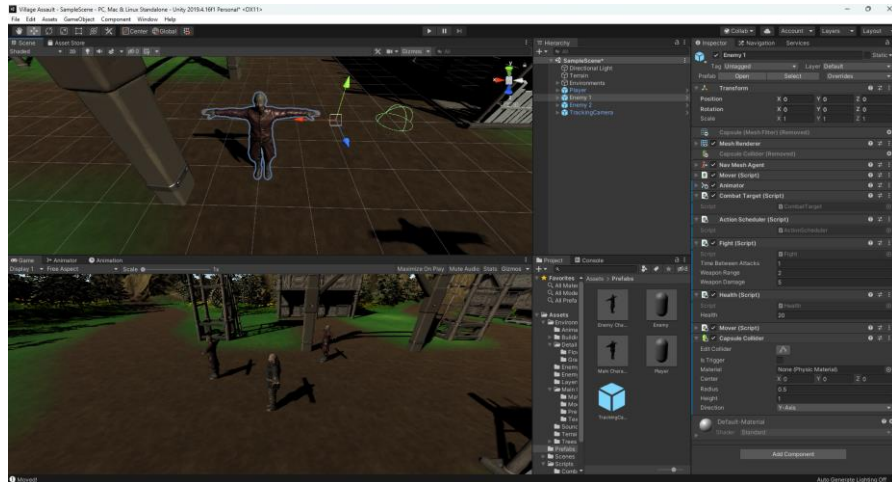
- Animations for stopping, walking, and running were assigned to the character using an animator. The character was merged with an existing cylinder object, which was made invisible by removing its Mesh Filter component. Then, the `UpdateAnimator()` method was created, and the speed value was matched with "forwardSpeed" using `vector3`.



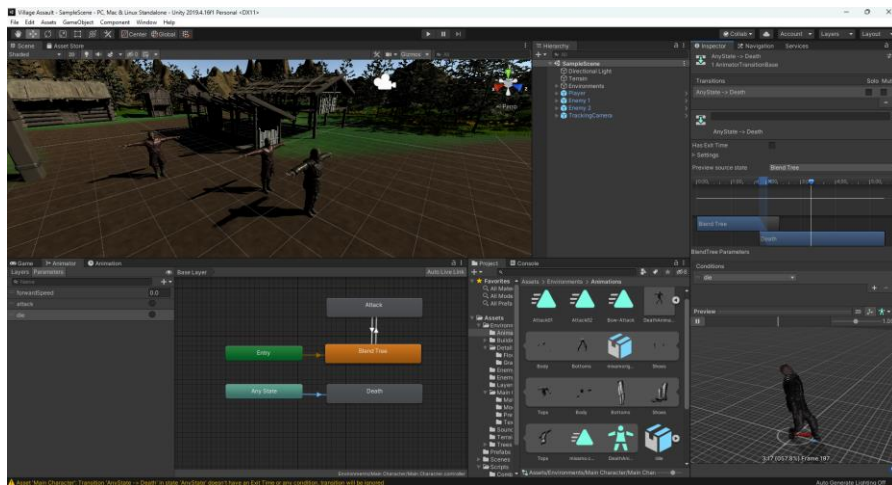
Attack System:

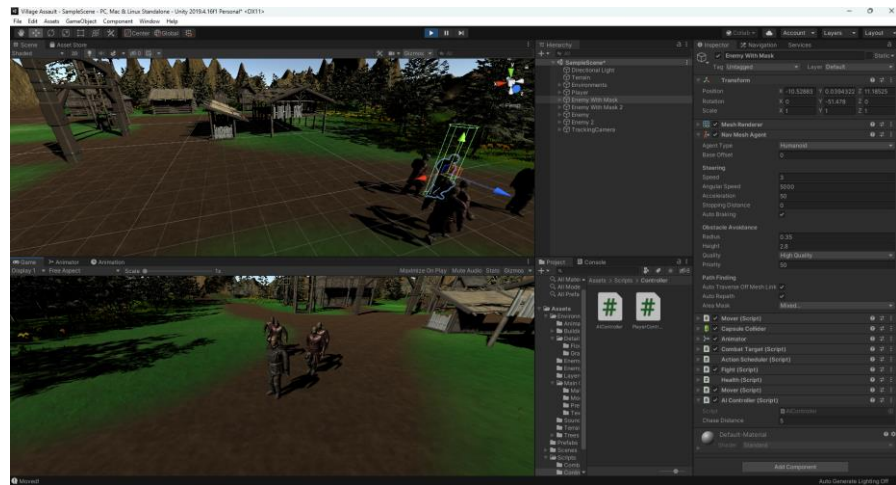
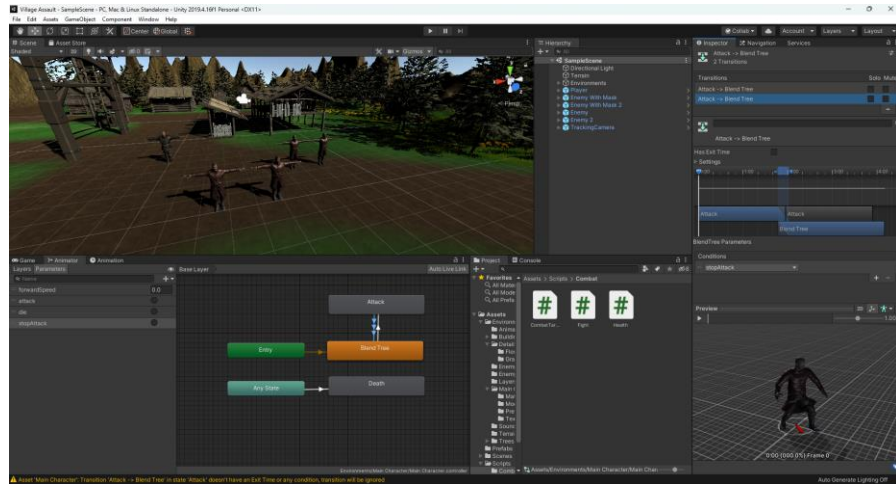
- Animator controllers were used to set idle, walk, and run animations for the character. The characters were combined with a cylinder object, which was made invisible by removing the Mesh Filter component to keep it from showing. The `UpdateAnimator()` method was set up to match the character's movement speed with the `forwardSpeed` parameter using `vector3`.





- An Attack System was set up. Needed codes and animations were prepared to make fighting possible. Triggers in the Blend Tree were arranged to start different attack animations during the game. Script files were made to handle these animations and make sure they work right in the game.





Note: Additionally, comment lines in the script code are made within the codes.

Tufan Şahin Düzel - 1002729